

Q3 2025 Earnings Call

Company Participants

- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, Chairman, President and Chief Executive Officer
- Satya Kumar, Corporate Vice President, Investor Relations and Treasury

Other Participants

- C.J. Muse, Cantor Fitzgerald
- Chris Caso, Wolfe Research
- Harlan Sur, J.P. Morgan
- Joseph Moore, Morgan Stanley
- Krish Sankar, TD Cowen
- Thomas O'Malley, Barclays
- Timothy Arcuri, UBS
- Vijay Rakesh, Mizuho Securities
- Vivek Arya, Bank of America Securities

Presentation

Operator

Thank you for standing by. Welcome to Micron's Third Quarter 2025 Financial Call.

At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions) As a reminder, today's program is being recorded.

And, now, I'd like to introduce your host for today's program, Satya Kumar, Investor Relations. Please go ahead, sir.

Satya Kumar {BIO 18241786 <GO>}

Thank you, and welcome to Micron Technology's fiscal third quarter 2025 financial conference call.

On the call with me today are Sanjay Mehrotra, our Chairman, President, and CEO; and Mark Murphy, our CFO.

Today's call is being webcast from our Investor Relations site at investors.micron.com, including audio and slides. In addition, the press release detailing our quarterly results has been posted on the website, along with the prepared remarks for this call.

Before we begin, let me remind everyone that today's discussion contains forward-looking statements that are subject to risks and uncertainties. These forward-looking statements include statements regarding our future financial and operating performance, including our guidance, as well as trends and expectations in our business, market, industry, and regulatory and other matters. These statements are based on our current assumptions, and we assume no obligation to update these statements. Please refer to our most recent financial report on Form 10-K and our other filings with SEC for more information on these risks and uncertainties that could cause actual results to differ materially from expectations.

Today's discussion of financial results is presented on a non-GAAP financial basis unless otherwise specified. The reconciliation of GAAP to non-GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on financial conferences that we may be attending. You can also follow us on LinkedIn, X, and YouTube.

I'll now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Satya. Good afternoon, everyone.

Micron's strong competitive position and solid execution delivered record revenue in fiscal Q3, with revenue, gross margin, and EPS all exceeding the high end of our guidance ranges. Data center revenue more than doubled year-over-year and reached a record level and consumer-oriented markets had strong sequential growth. We generated substantial free cash flow in the quarter, even as we continue to make strategic investments critical to sustain long-term growth. I'm thankful to all our Micron team members for their focus and execution, which made these results possible.

In fiscal Q3, DRAM revenue reached a new record, driven by a nearly 50% sequential growth in HBM revenue. We remain the sole supplier in volume production of LP DRAM in the data center. In NAND, we achieved a new quarterly record for market share across data center SSDs, as well as client SSDs in calendar Q1. For the first time ever, during calendar Q1, Micron has become the number-two brand by share in data center SSDs, according to third-party data.

Looking ahead to fiscal Q4, we see a robust demand environment and expect to grow revenue by 15% sequentially to a record \$10.7 billion at guidance midpoint. In June, we have completed a strategic reorganization of our business units around key market segments to capitalize on the tremendous AI growth opportunity ahead. As high-performance memory and storage becomes increasingly critical to enabling AI-driven innovation, this new structure enhances Micron's ability to engage more deeply with customers by shifting more resources to AI-focused opportunities across our portfolio.

We are making excellent progress on our 1-gamma DRAM technology node, with yield damping ahead of the record pace we have achieved on our 1-beta node. We completed several key product milestones during the quarter, including the first qualification sample shipments of 1-gamma-based LP5 DRAM.

Micron 1-gamma DRAM leverages EUV, and the node provides a 30% improvement in bit density, more than 20% lower power, and up to 15% higher performance compared to 1-beta DRAM. We will leverage 1-gamma across our entire DRAM product portfolio to benefit from this leadership technology. In NAND, we reached a record high mix of QLC bits in the quarter. We started qualifications for new high-performance SSD products based on our G9 2-terabit QLC NAND, and we continue to ramp our G9 node at a pace consistent with demand.

We are making disciplined investments in our global operations network to add to supply in line with demand over time. Two weeks ago, with support from the Trump administration, Micron announced plans to invest approximately \$200 billion in the U.S., which includes \$150 billion in manufacturing and \$50 billion in R&D over the next 20-plus years.

As part of this \$200 billion investment plan, Micron plans to invest an additional \$30 billion beyond previously announced plans, which includes building a second leading-edge memory fab

in Boise, Idaho; expanding and modernizing our existing fab in Manassas, Virginia, serving the automotive, aerospace, defense, and industrial markets; and bringing advanced packaging capabilities to the U.S. to support our long-term HBM growth plans after we have established sufficient DRAM wafer scale in our U.S. operations.

We are pleased with the strong endorsement we received for our technology, products, and investment plans from our customers and ecosystem partners as part of this announcement.

Our first Idaho fab, ID1, achieved another key construction milestone in June. We expect first DRAM wafer output at ID1 to begin in the second half of calendar 2027, with customer qualifications to follow. The second Idaho fab, ID2, will benefit from manufacturing economies of scale with ID1 and add to R&D co-location benefits with greater efficiencies and faster time-to-market. To meet anticipated demand, ID2 will begin production before the first New York fab. We expect to begin ground preparation in New York later this year following the completion of state and federal environmental reviews.

Turning to our end markets. In data center, we expect the CY '25 server market to grow mid-single-digit percentage in units, largely driven by significant growth in AI servers. In fiscal Q3, data center DRAM revenue reached a new record for the fourth consecutive quarter, driven by strong growth and share gains in HBM and robust performance by our industry-leading portfolio of high-capacity DIMMs and low-power server DRAM products.

We are executing well on our HBM ramp and product development roadmap. Our yield and volume ramp on HBM3E 12-high is progressing extremely well, and we expect shipment crossover in FQ4. We expect to reach HBM shares similar to our overall DRAM shares sometime in the second half of calendar 2025.

At AMD's Advancing AI event earlier this month, we announced that Micron's HBM3E 36-gigabyte 12-high has been designed into AMD's Instinct MI355X GPU platform. We are now shipping HBM in high volume to four customers, spanning both GPU and ASIC platforms.

As generative-AI workloads grow in size and complexity, the performance demands on HBM continue to rise. Micron's HBM4 leverages our well-established 1-beta DRAM technology, along with an internally developed and manufactured advanced CMOS logic-based die, to deliver bandwidth exceeding 2 terabytes per second per memory stack, over 60% higher performance than the previous generation.

Additionally, Micron's HBM4 offers a 20% lower power consumption compared to the already industry-leading power performance on our HBM3E 12-high product, setting new benchmarks in power efficiency for this product category. The expanded interface for HBM4 facilitates rapid communication and a high-throughput design that accelerates the inference platform performance of large language models and chain-of-thought reasoning systems. Micron has delivered samples of HBM4 to multiple customers and expects to ramp volume production in calendar 2026, aligned with our customers' plans.

We are exceptionally well positioned for the ramp of HBM4. Building on the success of our HBM3E ramp, we have high-quality, field-proven technology and have executed a robust and significant ramp in our HBM manufacturing capacity. We have deep relationships with practically every major customer of HBM and have earned their trust with our execution, delivering the world's lowest-power, highest-performance HBM.

Our portfolio of high-capacity DIMMs and low-power server DRAM solutions delivered another record revenue quarter. Micron has pioneered the adoption of LP DRAM for servers, and we

continue to maintain our sole-source position for LP in server. Together, our high-capacity DIMM and LP server products have already generated multiple billions of dollars in revenue in fiscal 2025, reflecting a remarkable five-fold growth compared to the same period in the previous year.

During calendar Q1, for the third consecutive quarter, Micron achieved a record for data center SSD market share, driven by our portfolio of differentiated products enabled through vertical integration. In fiscal Q3, our data center 9550 performance SSD, which is on the NVIDIA GB200 NVL72 recommended vendor list, completed additional customer qualifications at multiple OEMs. Micron's 9550 SSDs provide an industry-leading performance and energy-efficient Gen5 data center storage solution for AI server systems. We continue to qualify additional customers and ramp revenue for our 6550 ION 60-terabyte capacity SSDs.

Turning to PC. We expect PC market units to grow in the low-single-digit percentage range in calendar 2025. In the quarters ahead, key catalysts for growth include the increasing adoption of AI-enabled PCs in the Windows 11 upgrade cycle. Micron is focused on bringing differentiated high-performance products to the PC market. Our strong SSD portfolio resulted in Micron achieving a record-high client SSD market share in calendar Q1. Tomorrow, we will be announcing our new G9 QLC 2-terabit-based SSD, featuring our proprietary Adaptive Write Technology, which enables 4x faster write performance. This technology expands the addressable market for QLC SSDs by delivering performance equivalent to TLC NAND for most consumer use cases.

Turning to mobile. We expect smartphone units to grow low-single-digits in calendar 2025. AI adoption remains a key driver of DRAM content growth for smartphones, and we expect more smartphone launches featuring 12 gigabyte or more compared to 8 gigabyte of capacity in the average smartphone today.

Micron is focused on providing solutions to the high-end smartphone segments, leveraging our leading 1-beta and 1-gamma technology nodes for LP5X DRAM and G8 and G9 technology nodes for our UFS4 NAND products. During the quarter, we began shipping qualification samples of the industry's first LP5X memory built on the 1-gamma node, offering a wide range of capacities and industry-leading speed grades for 2026 flagship smartphones.

Micron's 1-gamma LP5X DRAM is engineered to accelerate AI applications in high-end devices, delivering over 25% faster recommendations across several use cases, while reducing power consumption by 20%, all in an ultra-thin form factor ideal for mobile. In NAND, we secured a key customer design win and ramped high-volume production of our G9-based UFS4 products. The strength of our mobile portfolio was further recognized through top-quality awards from seven smartphone OEMs during the quarter.

Turning to automotive, industrial, and consumer-embedded markets, we expect increasing adoption of L2 and L3 ADAS features and AI-enabled in-vehicle infotainment systems to drive memory and storage content growth, as well as higher bandwidth requirements. Micron is positioned for long-term success in the automotive market, with new product introductions, such as the industry's first 1-beta dual-channel LP5 DRAM with high-speed 9.6-gigabits-per-second support, which achieved production-readiness during the quarter.

In industrial, we are seeing a resumption in our growth, as customers increase their investments for the adoption of AI, including in key areas like factory automation. Micron is driving price improvements with a market backdrop of constrained D4 and LP4 supply and low distributor channel inventory.

Now, turning to our market outlook. Customer inventory levels have been healthy overall across end markets, and there may have been some tariff-related pull-ins by certain customers. Our

customers continue to signal a constructive demand environment for the remainder of this calendar year, and we remain agile to adjust to any unforeseen demand changes that may occur due to macro conditions or the evolving tariff-related situation.

We expect CY '25 industry DRAM bit demand growth to be in the high-teens percentage range and industry NAND bit demand growth to be in the low-double-digit percentage range. We expect Micron's bit supply growth to be below industry bit demand growth for non-HBM DRAM and NAND. Over the medium term, we anticipate industry bit demand growth of mid-teens CAGR for both DRAM and NAND.

As previously communicated, our efficient node conversions will result in 10% structurally lower NAND wafer capacity ending fiscal 2025 versus the end of fiscal 2024 levels. Additionally, given NAND technology transitions provide a significant increase in overall bit output, Micron plans to manage our node conversions at a measured pace consistent with our demand.

Recent press reports have discussed the end-of-life of D4 and LP4 products. Micron's leading-edge DRAM nodes, such as 1-beta and 1-gamma, are focused on the latest generation products, such as D5, LP5, and HBM, and are not utilized to produce D4 and LP4. D4 and LP4 products are largely produced in our 1-alpha DRAM node. Micron had sent EOL notices for these products to customers in high-volume segments, like mobile, client, data center, and consumer, several months ago, with final shipments occurring in two to three quarters from now.

This EOL process is similar to prior transitions from one generation of memory to another, and consistent with history. Micron intends to support its longevity customers with long-term and relatively low-volume requirements in segments like automotive, industrial, defense, and networking with supply of these 1-alpha DRAM products for several years.

In the near term, customers in the high-volume segments are starting to see increasing shortages of D4 products. We are now on allocation for these products and are working with customers to try and support their high-priority, near-term demand. D4 revenues are low-single-digit percentage of our revenues in second half of fiscal 2025. We anticipate LP4 shortages may also increase as a result of EOL.

In closing, Micron's record Q3 revenue performance and strong Q4 outlook are the results of our strategic focus and consistent execution. As AI drives unprecedented demand for high-performance memory and storage, Micron is exceptionally well positioned to capitalize on this transformative era. Our leadership in technology, highlighted by progress in HBM, 1-gamma DRAM, and G9 NAND, alongside disciplined global manufacturing investments, supports our path to sustained growth.

We are confident that our strategic direction, innovation capabilities, and the execution by our exceptional team will continue to create meaningful value for our shareholders, customers, and employees. We are on track to deliver record revenue with solid profitability and free cash flow in fiscal year '25, while we invest to build on our leadership to addressing growing AI-driven memory demand.

I will now turn it over to Mark for our financial results and outlook.

Mark Murphy {BIO 16136048 <GO>}

Thank you, Sanjay, and good afternoon, everyone.

Micron delivered strong results in fiscal Q3, with revenue, gross margin, and EPS all above the high end of the guidance ranges provided in our last earnings call. Total fiscal Q3 revenue is \$9.3

billion, up 15% sequentially and up 37% year-over-year, and a quarterly revenue record for Micron. Higher sequential revenue was driven by growth across our end markets, including record data center revenues and strong sequential growth in consumer-oriented markets.

Fiscal Q3 DRAM revenue was \$7.1 billion, up 51% year-over-year, and represented 76% of total revenue. Sequentially, DRAM revenue increased 15%, with bit shipments increasing over 20%, and prices decreasing in the low-single-digit percentage range, primarily due to a higher consumer-oriented revenue mix.

Fiscal Q3 NAND revenue was \$2.2 billion, up 4% year-over-year, and represented 23% of Micron's total revenue. Sequentially, NAND revenue increased 16%, with bit shipments increasing in the mid-20s percentage range, and prices decreasing in the high-single-digit percentage range.

Now turning to revenue by business unit. Compute and Networking Business Unit revenue was \$5.1 billion, up 11% sequentially, and a quarterly record. This performance was driven by a nearly 50% sequential increase in HBM, along with growth in our high-capacity DRAM and low-power server DRAM.

Revenue for the Storage Business Unit was \$1.5 billion, up 4% sequentially. This growth was primarily driven by an increase in consumer-oriented revenue.

Mobile Business Unit revenue was \$1.6 billion, up 45% sequentially. Sequential revenue growth was due to reduced customer inventories and strong demand from DRAM content growth.

Embedded Business Unit revenue was \$1.2 billion, up 20% sequentially, supported by growth in industrial and consumer-embedded markets.

The consolidated gross margin for fiscal Q3 was 39%, up 110 basis points sequentially, and up 250 basis points versus the midpoint of our guidance. Gross margins were above the high end of our guidance range, primarily due to better prices for both DRAM and NAND, partially offset by a higher consumer-oriented mix.

Operating expenses in fiscal Q3 were \$1.1 billion, up \$87 million quarter-over-quarter and in line with our guidance range. The increase was primarily driven by higher R&D investments and labor-related costs.

We generated operating income of \$2.5 billion in fiscal Q3, resulting in an operating margin of 26.8%, up approximately 190 basis points sequentially, and up 13 percentage points year-over-year.

Fiscal Q3 taxes were \$306 million on an effective tax rate of 12.3%, lower than our guidance due to the effects of one-time discrete items in the quarter.

Non-GAAP diluted earnings for share in fiscal Q3 was \$1.91, above the high end of the guidance range, with 22% sequential growth and up over 200% versus the year-ago quarter.

Turning to cash flows and capital spending. In fiscal Q3, our operating cash flows were over \$4.6 billion and our capital expenditures were \$2.7 billion. Free cash flows in the quarter were over \$1.9 billion, the highest quarterly amount in over six years.

Ending inventory for fiscal Q3 was \$8.7 billion, or 139 days. Inventory was down \$280 million sequentially, and inventory days were down 19 days sequentially, driven by strong sequential bit shipment growth in both DRAM and NAND.

On the balance sheet, we held a record \$12.2 billion of cash investments at quarter end and maintained \$15.7 billion of liquidity when including our untapped credit facility. During fiscal Q3, we refinanced our \$900 million 2027 notes with \$1.7 billion in new notes maturing in fiscal years 2033 and 2036. We closed the quarter with \$15.5 billion of debt, maintaining low net leverage and a weighted-average debt maturity of 2032.

Now turning to our outlook for the fourth fiscal quarter. We expect our revenue growth to be weighted towards DRAM, supported by robust pricing execution, favorable product mix, and continued cost improvements, all of which benefit gross margins.

Operating expenses for fiscal Q4 are projected to be approximately \$1.2 billion, with a sequential increase primarily driven by planned R&D investments in future technology nodes and HBM product development.

Our fiscal 2025 capital spending plans remain unchanged at approximately \$14 billion. The overwhelming majority of the fiscal 2025 CapEx is to support HBM as well as facility, construction, backend manufacturing, and R&D investments.

We expect a fiscal Q4 tax rate of around 13%. As previously disclosed, our fiscal 2026 tax rate is expected to be in the high-teens percentage range following Singapore's adoption of the global minimum tax. Any impacts that occur due to potential new tariffs are not included in our guidance.

With all these factors in mind, our non-GAAP guidance for fiscal Q4 is as follows. We expect revenue to be \$10.7 billion, plus or minus \$300 million; gross margin to be in the range of 42%, plus or minus 100 basis points; and operating expenses to be approximately \$1.2 billion, plus or minus \$20 million. We expect our fiscal Q4 tax rate to be around 13%. Based on a share count of approximately 1.15 billion shares, we expect EPS to be \$2.50 per share, plus or minus \$0.15.

With another quarter of shipment growth forecasted in fiscal Q4, we expect to exit fiscal 2025 with tight DRAM inventories, significantly reduced NAND inventories, and overall company DIO near our target levels. With low inventories on hand and a constructive demand environment, we will continue to focus on improving pricing and further strengthening our product mix.

In fiscal Q3, Micron delivered earnings above the guidance range, achieved record revenue, and continued ramping our industry-leading HBM. We also began transitioning to a new market segment-focused business unit structure. Starting in fiscal Q4, we will report revenue, gross margin, and operating margin metrics across these new business units. With strong execution and a differentiated product portfolio, Micron is well positioned to maintain our leadership and to deliver record revenue and significantly improved profitability once again in fiscal Q4.

Thank you for joining us today. We will now open up for questions.

Questions And Answers

Operator

(Question And Answer)

Certainly. (Operator Instructions) And our first question comes from the line of Timothy Arcuri from UBS. Your question please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. Sanjay, I wanted to ask sort of how you see the HBM TAM scaling with the accelerator TAM. One thing that's been clear over the past three months or so is that the accelerator TAM is definitely upscaling and growing a lot. And it seems, if I take their projections and I take your HBM projections, it kind of seems like the HBM market is going to remain 15% to 20% of what the overall accelerator market is. I guess how do you see the HBM TAM scaling with that market? And then is there some sort of limit or like asymptote that we begin to reach in terms of HBM attached to these GPUs and these custom ASICs?

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So with respect to HBM growth, we certainly see that HBM demand grows in the future. I'll tell you that in this year, if you look at calendar year '25, HBM is growing from last year about \$18 billion in revenue to approximately \$35 billion in calendar year '25.

We see in calendar year '26, if you look at HBM bit demand growth, it will significantly exceed the overall DRAM industry demand growth. And, of course, the transition from 8-high to 12-high in HBM in '26, transition to HBM4 as well, which is even a higher value product, all of that essentially bodes well in terms of the value of HBM and the overall need for HBM in these accelerator platforms that are, of course, evolving fast. And, of course, our customers are very much focused on working with us and looking at their platforms, assessing the mix of 12-high and HBM4.

And then when we look at '27, that goes into HBM4E. '28 starts introducing customization in HBM as well. So the trajectory of HBM in terms of the value proposition in these accelerators continues to be strong and healthy. And, of course, we are well positioned. As we shared with you, we have already begun to sample our HBM4 products. And, again, we plan to maintain our leadership with respect to specifications here. And we have certainly demonstrated a capacity ramp up as well.

So, we remain excited about HBM. We see this as clearly a strong growth driver as well as value driver for our business.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Thanks a lot, Sanjay.

Operator

Thank you. And our next question comes from the line of Vivek Arya from Bank of America Securities. Your question please.

Q - Vivek Arya {[BIO 6781604](#) <GO>}

Thanks for taking my question. I wanted to talk about gross margins. So, first, what is driving the upside sequentially? And then is this the new baseline for gross margins? And then what would be kind of the puts and takes as we look at the next few quarters beyond Q4? Thank you.

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yes, Vivek, it's Mark. So, in the third quarter, I think maybe start with what happened in gross margin in the third quarter. We had a strong sequential volume growth in the quarter, but versus the guide, the defining factor was we had -- prices were down, as I said in the opening comments, but we had better-than-expected pricing that drove the margin improvement versus expectations. I'll add to that we had good cost performance and all, but certainly the price was better than we had expected.

Now that carries on into re-baselining on the fourth quarter. So we provided an indication of the fourth quarter before when the market was in transition. And we said at that time that the inventory outlook that we would -- it was improving when it had weakened a bit late last year. And we had intended to work to inflect price. And, as you know, a few weeks later, the situation with the Liberation Day tariffs, the environment got a bit more challenging.

The condition of the market is better than expected since then, and it's strengthened through the quarter. You saw that outperformance. And then we've executed well. And so what you see here from this re-baseline of the business, you see us add to that through favorable mix Q3 to Q4. That's more DRAM growth than NAND. So, you have favorable mix there. And then more data center than consumer-oriented mix, which was weighing down the margins in the third quarter. So, it's -- again, it's a continued good market backdrop. We're going to continue to focus on pricing, but third quarter to fourth quarter we have favorable mix effects that is helping drive margins up to the current guide of 42%.

Q - Vivek Arya {[BIO 6781604](#) <GO>}

And anything beyond this, Mark? Any puts and takes as we look at the next several quarters? Is this kind of the new baseline or is it going to be entirely dependent on the direction of pricing?

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yes. We're not going to provide Q1 guidance. We are positive on the trajectory of the business. The market environment does remain constructive, particularly in DRAM versus NAND. Again, we're focused on pricing and making sure to put our bits in the right places. As we said in the prepared remarks, our inventories are very tight, particularly on leading edge and then now even pockets of some of the old legacy and DRAM. So we're going to have some bit constraints going into first quarter, but we're going to mix to DRAM, higher-value DRAM and higher-value NAND products. And so we think gross margins can be up.

Q - Vivek Arya {[BIO 6781604](#) <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of C.J. Muse from Cantor Fitzgerald. Your question please?

Q - C.J. Muse {[BIO 6507553](#) <GO>}

Yes. Good afternoon. Thank you for taking the question. I was hoping to revisit your comments on HBM. So you're talking 23%, 24% market share, and it sounds like you're now calling not exiting the year, but sometime in the second half. So that's coming in better. And so I guess, first question is, should we be thinking that that number is kind of a 2, 2 to 3, depending, I guess, on seasonality of the \$35 billion number? And then perhaps more importantly, as you think about HBM growth in calendar '26, how should we be thinking about the contributions to that growth, both from bits as well as higher ASPs as you go to next-generation products? Thanks so much.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

C.J., I didn't quite understand the question around 2, 2 to 3. What exactly did you say there?

Q - C.J. Muse {[BIO 6507553](#) <GO>}

Just trying to understand how you talked about getting to your DRAM market share, which I think you said 23%, 24% for the HBM market. And your previous comments were exiting the year. Now, you're saying sometime in the second half. So trying to get an idea of kind of what that revenue number looks like.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

We are already at, if you look at our performance, based on FQ3, we are already at more than \$6 billion run rate here with our HBM, and certainly continuing to ramp up our HBM output, shifting production to 12-high HBM as well.

So, yes, it could be that compared to what we said before in terms of toward the end of the calendar year, we expect to get to our HBM share in line with our DRAM share. So, versus that, it could be that we do end up getting to our industry share for -- in line with industry share for HBM share earlier than that. So it could be. And that's really absolutely based on our very, very strong execution, and strong execution in terms of output, in terms of yield ramp.

We noted that our 12-high yield ramp is actually going faster than our 8-high yield ramp. Of course, built on all the learning that we have had, the team has done an excellent job with capacity ramp as well. All of that is contributing to our strong performance on HBM.

So we are extremely pleased with where we are and extremely pleased with not just execution today and our ability to ramp up yield and capacity successfully and build in the process trust with the customers, but very pleased with the roadmap that is ahead for us for HBM as well. So I mean, these are things that will obviously play an important role in calendar year '26 as we work closely with the customers to understand their overall mix requirements for 2026.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Thank you.

Operator

Thank you. And our next question comes from the line of Thomas O'Malley from Barclays. Your question please.

Q - Thomas O'Malley

Hey, guys. Thanks for taking the questions. Two here. So my first on the HBM side, you're obviously reaching your share target a little bit earlier. Going into next year, are you guys ready to talk about what you think that your normalized share will be? Obviously, there's difficulties with some of your competitors getting qualified. If that continues into next year, do you guys have a view of if you're able to increase capacity, and if you are, what that share may look like?

And then two, on the NAND side, where does utilization stand today? And, obviously, you're going through some bit transitions right now internally, which has allowed you to kind of keep some things offline. But, as you move into next year, what is your plan for utilization? Is that going to be market-dependent, or at some point, do you need to start bringing on utilization just from the fact that there's a gross margin headwind? Thank you.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So, as I said, we are very pleased that we are executing well, and with what we told you several quarters ago in terms of our share objectives with HBM, that we are going to be able to achieve

that goal for this year in second half of '26.

Now our HBM is really at scale. It has healthy shares, and we are successfully delivering. So, going forward, of course, this becomes like part of our overall product portfolio. And just like we manage the rest of the portfolio with respect to total eye on ROI and profitability, and, of course, HBM really positions us very well with respect to our profitability objectives, we will, of course, continue to manage the mix of our products in the portfolio, including that of HBM, including the share that sometimes can move around in different parts of the end market segments that we address.

So I think that's how you have to look at it. Overall, really very pleased with our HBM products, 12-high as well as HBM4, built with our well-proven 1-beta technology, which also gives us cost-effective benefits. And, of course, the packaging technology well-established and company, of course, has been investing in expanding our HBM back-end capacity as well.

We have talked about investments in Singapore, bringing assembly and packaging capacity there in line, starting production, targeting that for 2027 timeframe. And I'll tell you that HBM, given that it uses our well-proven 1-beta technology, is really fungible capacity with the rest of the portfolio as well. So this gives us plenty of flexibility in terms of managing our business and extremely focused on really meeting customers' requirements and continuing to deliver and enhance our product capabilities to meet their growing requirements.

And your second question around NAND in terms of utilizations, so we have said before that, toward the end of fiscal '25, our NAND overall capacity would be down versus end of fiscal '24 by about 10%. And that's a structural reduction in capacity. And, of course, as we have discussed before, that structural reduction was implemented to achieve capital-efficient next node transition for us, like G9 node transition. So, as that capacity has come down, of course, our underutilization has come down as well, although part of NAND continues to remain underutilized. I must note here that, of course, leading edge of NAND is fully utilized.

Q - Thomas O'Malley

Thank you, Sanjay.

Operator

Thank you. And our next question comes from the line of Harlan Sur from J.P. Morgan. Your question please.

Q - Harlan Sur {BIO 6539622 <GO>}

Hey, good afternoon. Thanks for taking my question. And if I rewind back one year ago, during your June earnings call, you guys did say back then that you were sold out on your HBM supply through calendar '25, and that majority of that committed supply was locked in from a pricing perspective. One year later, where are you on your negotiations for your calendar '26 HBM supply and pricing discussions? Is the team supply outlook for next year fully committed to or maybe a better way to frame it, right, because I know that the HBM3E and HBM4 12-high qualification cycles might be taking a bit longer. But maybe the other way to frame it is if you look at your customers' calendar '26 forecasted demand for HBM, is it exceeding your forecasted supply capability?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

With respect to HBM in 2025, yes, very pleased that what we told you back a year ago, we are continuing to deliver on that, as I said earlier. And, yes, our HBM is sold out for 2025. And, as I mentioned earlier, we are working closely with our customers as their platforms, AI-accelerated platforms, both with GPUs as well as with ASICs are continuing to evolve and move fast. And they themselves are working on overall their supply chain requirements and product mix with respect to HBM3E 12-high as well as HBM4.

So we continue to work with the customers in those. And we are still in the middle of '25 and for 2026, as I said earlier, we see, with respect to bit demand growth, strong trend in 2026, bit demand growth for HBM significantly exceeding DRAM demand growth. And, of course, we have a strong product roadmap. And we are working on continuing, as we said, we sampled HBM4, focused on getting these products qualified with the customers. And HBM3E 12-high is already in volume productions and doing very well with our yield ramp there as well. So very much focused on addressing the '26 needs for the customers and executing well and remaining extremely focused on our execution for 2026.

Q - Harlan Sur {[BIO 6539622 <GO>](#)}

Thanks, Sanjay.

Operator

Thank you. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question please.

Q - Joseph Moore {[BIO 17644779 <GO>](#)}

Thank you. I wonder if you could talk in terms of your long-term CapEx plans and the sort of 35% of revenue levels. Is that kind of a reasonable guideline to think about for fiscal '26, or are you thinking because of the opportunity here, you need to invest ahead of that? Thank you.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes, Joe, I mean, we have a generational tech transition opportunity in front of us that Micron is exceptionally well positioned for. And we've talked about inventory levels for some time now, getting leaner, especially, being very tight on the leading edge. And with the silicon requirements of HBM and the trade ratio, we need to build greenfield capacity. So that's underway. If you look at our CapEx forecast, we had a little bit lower CapEx than we expected in the quarter at \$2.7 billion, but you'll see by the guide that we're going to see that CapEx increase in the fourth quarter. So we stuck with our approximately \$14 billion number.

I think, as we go forward, we're continuing to build out the greenfield capacity and there'll be equipment installs on the new nodes. There's grants involved in construction. So the timing of that spend can be a bit lumpy, but we anticipate generating free cash flow in the fourth quarter. And you will see we'll just continue to build out -- build our capacity.

I do want to note that we continue to make steps to improve the balance sheet further in the quarter. We're down -- net debt now down to \$3 billion. We further reduce maturities in the short end of the maturity schedule, taking out the \$900 million '27 notes and issuing \$1.75 billion further out. And we continue to be in a great position with a flexible balance sheet to invest in the business, make sure we maintain technology leadership, and then return capital to shareholders through dividend and opportunistic buyback.

Q - Joseph Moore {[BIO 17644779 <GO>](#)}

Great. Thank you. Great quarter.

Operator

Thank you. And our next question comes from the line of Krish Sankar from TD Cowen. Your question please.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Yes. Hi. Thanks for taking my question. I have two questions for Sanjay, or two-part question. One is, when you look into HBM4, given the double I/O count and through-silicon vias, what kind of trade ratio should we expect for HBM4? And also, when you look at -- broadly speaking, is there a difference in HBM bits and margin profile between selling to GPU versus ASIC customers? Thank you.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

With respect to HBM4, its trade ratio is greater than 3. For the HBM3E, previously we have shared that trade ratio is approximately 3, and HBM4 has a higher trade ratio than 3. It has a larger die size, and, of course, as you noted, it has higher performance. We shared 60% higher performance, given the high bandwidth interface that this has. And HBM4E will have a ratio that would be even greater.

So as we have shared with you in the past, that as we are going from HBM3E to 4 to 4E, the trade ratio is going from about 3 towards 4 in that time frame. So, of course, this puts -- HBM trade ratio and the growth of HBM absolutely puts a pressure on the non-HBM supply in the industry as well.

And with respect to your question on the margin difference for GPU versus ASIC, of course, HBM commands high value, whether it's in GPU accelerators or ASIC-based accelerators. And that value proposition is only growing. When you look ahead over the years, the content with the future generation GPUs or ASIC platforms that will be out there of HBM continues to -- is expected to continue to increase as well.

We have shared with you in the past that going from about 200 gigabytes to 288 gigabytes in GPU accelerators today, as well as in the ASIC-based accelerators in that range, going to higher levels as we go from 8-high to 12-high capacity. So we don't -- so value proposition is strong, both for GPU accelerators as well as ASIC accelerators. And, of course, we don't get into the specifics or differentiating the two here.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Thanks, Sanjay.

Operator

Thank you. And our next question comes from the line of Chris Caso from Wolfe Research. Your question please.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Yes. Thank you. Just a clarification on one of the things you said in the prepared remarks. You talked about there may have been some tariff-related pull-ins by certain customers. I think, overall, your comments suggest that you -- at least my interpretation is that you think that the customer inventory levels are getting a bit cleaned up here. Could you clarify the remarks a little bit, put it in context of what you expect that means for customer inventory levels and how that kind of affects your view of bit demand in the second half?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Yes. As we said, customer inventory levels have been healthy overall across our end markets. Some customers may have some level of tariff-related pull-ins. We think the impact of that is relatively modest here, and customers definitely continue to signal a constructive demand environment for the remainder of the calendar year.

And, of course, the demand trends are driven by strength in AI-driven data center demand, demand coming back in the automotive, industrial, resumption of growth there, as well as distribution channels, and as we have discussed, smartphone and PCs, really, as AI penetration increases, the content and key stories there intact for AI smartphones as well as PC, we provided some details in our script there.

So we see a constructive demand environment as our customers discuss with us. And, again, the effect of the pull-in is relatively modest here. Our growth in Q3 as well as our exceptional record for FQ4 guidance is really driven by Micron's strong execution in the growing market for AI.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you.

Operator

Thank you. And our final question for today comes from the line of Vijay Rakesh from Mizuho. Your question please.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yes. Hi, Sanjay and Mark. Just a quick question on the HBM4. Just wondering how the qualification is progressing? And if you continue to see that similar power performance leadership that you guys have had on HBM3E, do you see that continuing on the HBM4 side as well? Thanks.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So we have provided early units to our customers. We have sampled the products to our multiple customers with HBM4, really very pleased with the execution and all the specs that we see our HBM4 delivering. As I mentioned, not only performance, but we plan to continue to focus on that power, a strong power position that we -- which is critically important in AI applications that we have established with our HBM portfolio as well.

So, of course, the qualifications, the customer qualifications are ahead of us, and we will be ready to meet customers' demand in 2026 timeframe. So HBM in terms of volume ramp is a 2026 product, and meeting the timeline requirements of customers with their next-generation platform that will be implementing the HBM4 products.

I want to again remind you that HBM4 users are well-proven, cost-effective, 1-beta technology node. And, of course, it has internal advanced CMOS logic die internally developed and internally

manufactured advanced CMOS logic die which positions us well as well and all of our experience of HBM3E ramp-up, where going from HBM3E 12-high to HBM3E -- going from -- sorry -- going from HBM3E 8-high to HBM3E 12-high, we told you that that ramp is going extremely well with our yields ahead of our plans, output ahead of our plans as well as yields faster than the ramp that we had experienced in 8-high. All of that gives us confidence and positions as well for HBM4 ramp as well. So, we feel very, very good about our overall capabilities to address the HBM4 markets in 2026.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

Got it. And, on the SSD side, just quickly, with your focus on AI servers, are you seeing an accelerated pull-through on the AI server side for SSDs as well? Thanks.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So we had mentioned on the data center side in SSDs in late Q4 of calendar '24 as well as early part of calendar '25, there had been some inventory digestion for the data center SSDs given that prior to this period that I just mentioned, there was a very strong demand growth for SSDs.

And, as we look ahead, we think second half of calendar '25 will be better than first half of '24 -- I mean first half of '25, which was impacted by some of that inventory digestion as well. And our data center SSDs are well positioned with some of these recently announced accelerator platforms as well. And that, of course, will be contributing toward the data center SSD future growth as well.

We are very pleased with our record data center SSD share as achieved in CQ1 for the third-party reports. And we are now a clear number-two brand in terms of market share at the end of CQ1. So we plan to continue to leverage our SSD portfolio to continue to focus on shifting the mix of the products toward higher-value parts of the NAND market.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

Great. Thank you.

Operator

Thank you. This does conclude the question-and-answer session as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.

This transcript may not be 100 percent accurate and may contain misspellings and other inaccuracies. This transcript is provided "as is", without express or implied warranties of any kind. Bloomberg retains all rights to this transcript and provides it solely for your personal, non-commercial use. Bloomberg, its suppliers and third-party agents shall have no liability for errors in this transcript or for lost profits, losses, or direct, indirect, incidental, consequential, special or punitive damages in connection with the furnishing, performance or use of such transcript. Neither the information nor any opinion expressed in this transcript constitutes a solicitation of the purchase or sale of securities or commodities. Any opinion expressed in the transcript does not necessarily reflect the views of Bloomberg LP. © COPYRIGHT 2026, BLOOMBERG LP. All rights reserved. Any reproduction, redistribution or retransmission is expressly prohibited.